Table S11. Exploratory single-target controls

A. Single-target-active versus neither

Target pair	Score target	Positive class	n positive	n negative	AUROC (95% CI)	
EGFR/HER2	EGFR	A-only	37	12	0.5766 [0.3761, 0.7680]	
EGFR/HER2	HER2	B-only	32	12	0.8672 [0.7109, 0.9818]	
JAK1/JAK2	JAK1	A-only	32	14	0.6049 [0.4241, 0.7790]	
JAK1/JAK2	JAK2	B-only	32	14	0.5446 [0.3504, 0.7388]	
JAK1/TYK2	JAK1	A-only	32	14	0.7455 [0.5558, 0.9085]	
JAK1/TYK2	TYK2	B-only	32	14	0.7522 [0.5804, 0.8996]	
PIK3CA/mTOR	PIK3CA	A-only	32	14	0.5357 [0.3549, 0.7121]	
PIK3CA/mTOR	mTOR	B-only	32	14	0.6183 [0.4576, 0.7768]	
AChE/BChE	AChE	A-only	25	15	0.6213 [0.4427, 0.7867]	
AChE/BChE	BChE	B-only	28	15	0.6071 [0.4310, 0.7786]	
F2/F10	F2	A-only	32	14	0.6339 [0.4621, 0.7879]	
F2/F10	F10	B-only	32	14	0.6406 [0.4353, 0.8304]	
PPARG/PPARA	PPARG	A-only	32	14	0.6685 [0.4631, 0.8504]	
PPARG/PPARA	PPARA	B-only	32	14	0.6384 [0.4330, 0.8281]	
PPARA/PPARD	PPARA	A-only	32	14	0.5223 [0.3303, 0.7143]	
PPARA/PPARD	PPARD	B-only	32	14	0.7009 [0.5290, 0.8527]	

S1: negative Vina docking energy, with higher values favorable (search seed 42). The positive class is the single-target class active at the scored target. Neither ligands are negative in panel A; the opposite single-target class is negative in panel B. Dual ligands are excluded. Counts include eligible observations with a valid score at that target. CI: 95% class-stratified percentile bootstrap, 10,000 resamples, seed 271828; linear 2.5th and 97.5th percentiles. Positive resamples are shared between the two comparisons. Score ties count as 0.5. Intervals are exploratory and unadjusted for multiplicity.

Table S11. Exploratory single-target controls

B. Single-target-active versus opposite single-target-active

Target pair	Score target	Positive class	n positive	n negative	AUROC (95% CI)	
EGFR/HER2	EGFR	A-only	37	32	0.1807 [0.0845, 0.2914]	
EGFR/HER2	HER2	B-only	32	37	0.7939 [0.6748, 0.8970]	
JAK1/JAK2	JAK1	A-only	32	32	0.6250 [0.4844, 0.7588]	
JAK1/JAK2	JAK2	B-only	32	32	0.3755 [0.2417, 0.5186]	
JAK1/TYK2	JAK1	A-only	32	32	0.3125 [0.1865, 0.4512]	
JAK1/TYK2	TYK2	B-only	32	32	0.6777 [0.5410, 0.8057]	
PIK3CA/mTOR	PIK3CA	A-only	32	32	0.4268 [0.2900, 0.5713]	
PIK3CA/mTOR	mTOR	B-only	32	32	0.6348 [0.4941, 0.7686]	
AChE/BChE	AChE	A-only	25	28	0.7000 [0.5500, 0.8286]	
AChE/BChE	BChE	B-only	28	25	0.5293 [0.3629, 0.6900]	
F2/F10	F2	A-only	32	32	0.4619 [0.3213, 0.6084]	
F2/F10	F10	B-only	32	32	0.5039 [0.3584, 0.6445]	
PPARG/PPARA	PPARG	A-only	32	32	0.5044 [0.3584, 0.6475]	
PPARG/PPARA	PPARA	B-only	32	32	0.5083 [0.3623, 0.6523]	
PPARA/PPARD	PPARA	A-only	32	32	0.5542 [0.4102, 0.6958]	
PPARA/PPARD	PPARD	B-only	32	32	0.7144 [0.5801, 0.8345]	

S1: negative Vina docking energy, with higher values favorable (search seed 42). The positive class is the single-target class active at the scored target. Neither ligands are negative in panel A; the opposite single-target class is negative in panel B. Dual ligands are excluded. Counts include eligible observations with a valid score at that target. CI: 95% class-stratified percentile bootstrap, 10,000 resamples, seed 271828; linear 2.5th and 97.5th percentiles. Positive resamples are shared between the two comparisons. Score ties count as 0.5. Intervals are exploratory and unadjusted for multiplicity.